ASTROSIGHT | PLANETARY SURFACE INTELLIGENCE

AUTONOMOUS LUNAR & MARTIAN TOPOGRAPHICAL CRATER ANALYSIS REPORT

Analyst / Researcher:	Dr. Helena Vance	**Target Body:**	**MOON**
Timestamp:	2026-10-07 12:25:17	**Resolution (GSD):**	10.0 m/px
Validation Status:	**VALID_CRATER_DETECTED**	**Craters Identified:**	2
Model Pipeline:	AstroSight-CraterNet-v2.4	**Processing Time:**	138 ms

1. TOPOGRAPHICAL DETECTION VISUALIZATION



2. CRATER FEATURE COORDINATES (X, Y)

ID	Feature Name	X (px)	Y (px)	Radius (px)	Diameter (km)	Confidence
crater_01	CRATER 01	421.0	286.0	40.0	0.8 km	97.8%
crater_02	CRATER 02	735.0	512.0	55.0	1.1 km	94.2%

3. EUCLIDEAN & REAL-WORLD SPATIAL DISTANCE MATRIX

Vector	Origin Crater	Target Crater	Pixel Dist (px)	Real Distance (m)	Real Distance (km)
VEC-01	CRATER 01	CRATER 02	386.88 px	3868.8 m	3.87 km

4. SCIENTIFIC SPECIFICATIONS & UN SDG 9

Mathematical Formulation: Planar Euclidean metric d = √((x■ - x■)² + (y■ - y■)²) scaled via orthorectified ground sampling distance 10.0 m/px. Real-world distance D = d × resolution.

United Nations SDG 9 Alignment: AstroSight advances UN Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by establishing open AI infrastructure for planetary remote sensing, crater topology, and deep-space mission research.

Scientific Disclaimer: *AstroSight automated crater boundary coordinates and distance vectors are generated for scientific research and mission planning. Topographical scale assumes standard planar projection for small-area orbital imagery. Always cross-correlate with high-resolution LROC NAC / HiRISE elevation models.*